Rajshahi University of Engineering & Technology
Department of Computer Science & Engineering
CSE 3102, Lab No. 10, Lab Work

****Create a database named "AI_DB"**

```
CREATE DATABASE AI_DB;
```

****Create two tables ai_developer and ai respectively.**

Solution:

<u>Table Schemas:</u>

```
CREATE TABLE ai_developer (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  country VARCHAR(100),
  website VARCHAR(150)
);


CREATE TABLE ai (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  domain VARCHAR(100),
  release_year INT,
  language_supported VARCHAR(100),
  open_source BOOLEAN DEFAULT FALSE,
  version VARCHAR(20),
  website VARCHAR(150),
  category ENUM('NLP', 'CV', 'RL', 'AGI', 'Other') DEFAULT 'Other',
  developer_id INT,
  FOREIGN KEY (developer_id) REFERENCES ai_developer(id)
);
```

Question: Populate the ai_developer table with several rows

Solution:

```sql
INSERT INTO ai_developer (name, country, website) VALUES
('OpenAI', 'USA', 'https://openai.com'),
('Google AI', 'USA', 'https://ai.google'),
('DeepMind', 'UK', 'https://deepmind.com'),
('Meta AI', 'USA', 'https://ai.facebook.com'),
('Stability AI', 'UK', 'https://stability.ai'),
('EleutherAI', 'Global', 'https://www.eleuther.ai'),
('Anthropic', 'USA', 'https://www.anthropic.com'),
('Microsoft Research', 'USA', 'https://www.microsoft.com/en-us/research'),
('Amazon AI', 'USA', 'https://aws.amazon.com/machine-learning'),
('Midjourney Inc.', 'USA', 'https://www.midjourney.com'),
('DeepSeek AI', 'China', 'https://deepseek.com'),
('xAI', 'USA', 'https://x.ai'),
('Alibaba Cloud', 'China', 'https://www.alibabacloud.com'),
('Perplexity AI', 'USA', 'https://perplexity.ai'),
('Google', 'USA', 'https://ai.google/gemini'),
('Manus AI', 'China', 'https://manus.im');
```

Question: Populate the AI table with several rows

Solution:

```sql
INSERT INTO ai (name, domain, release_year, language_supported, open_source, version,
website, category, developer_id) VALUES
('ChatGPT', 'Conversational AI', 2022, 'English, Multilingual', FALSE, '4.0',
'https://chat.openai.com', 'NLP', 1),
('BERT', 'Text Embedding', 2018, 'Multilingual', TRUE, '1.0',
'https://github.com/google-research/bert', 'NLP', 2),
('AlphaGo', 'Game Playing', 2016, 'N/A', FALSE, '1.0', 'https://deepmind.com', 'RL', 3),
('DALL·E', 'Text to Image', 2021, 'English', FALSE, '3.0', 'https://openai.com/dall-e', 'CV', 1),
('Stable Diffusion', 'Image Generation', 2022, 'English', TRUE, '2.1', 'https://stability.ai', 'CV', 5),
('Whisper', 'Speech Recognition', 2022, 'Multilingual', TRUE, '1.0',
'https://github.com/openai/whisper', 'NLP', 1),
('GPT-NeoX', 'Text Generation', 2022, 'English', TRUE, '20B', 'https://www.eleuther.ai', 'NLP', 6),
('Claude', 'Constitutional AI', 2023, 'English', FALSE, '2.0', 'https://www.anthropic.com', 'NLP', 7),
('Amazon Lex', 'Conversational Bots', 2017, 'Multilingual', FALSE, '1.0',
'https://aws.amazon.com/lex', 'NLP', 9),
('Midjourney', 'Generative Art', 2022, 'English', FALSE, 'v6', 'https://www.midjourney.com', 'CV',
10),
('DeepSeek-R1', 'Reasoning LLM', 2025, 'English, Chinese', TRUE, 'R1', 'https://deepseek.com',
'Other', 11),
```

('Grok', 'Conversational AI with real-time web', 2023, 'English', FALSE, '3.0', 'https://x.ai', 'NLP', 12),
('Qwen 3', 'Multimodal LLM', 2025, 'Multilingual', TRUE, '3.0', 'https://chat.qwen.ai', 'NLP', 13),
('Perplexity', 'Search and QA', 2022, 'English', FALSE, '1.0', 'https://perplexity.ai', 'NLP', 14),
('Gemini 2.5', 'Multimodal reasoning model', 2025, 'English', FALSE, '2.5 Pro Experimental', 'https://ai.google/gemini', 'AGI', 2),
('Manus AI', 'Autonomous agent', 2025, 'English', FALSE, '1.0', 'https://manus.im', 'AGI', 16);

Queries:

1. List every AI model along with its developer's name, ordered by release year descending.

```
SELECT
 a.name   AS model_name,
 d.name   AS developer_name,
 a.release_year
FROM ai AS a
JOIN ai_developer AS d
 ON a.developer_id = d.id
ORDER BY a.release_year DESC;
```

Result:

model_name	developer_name	release_year
Filter...	Filter...	Filter...
DeepSeek-R1	DeepSeek AI	2025
Qwen 3	Alibaba Cloud	2025
Gemini 2.5	Google AI	2025
Manus AI	Manus AI	2025
Claude	Anthropic	2023
Grok	xAI	2023
ChatGPT	OpenAI	2022
Stable Diffusion	Stability AI	2022
Whisper	OpenAI	2022
GPT-NeoX	EleutherAI	2022
Midjourney	Midjourney Inc.	2022
Perplexity	Perplexity AI	2022
DALL·E	OpenAI	2021
BERT	Google AI	2018
Amazon Lex	Amazon AI	2017
AlphaGo	DeepMind	2016

2. Count how many models each developer has built

```
SELECT
 d.name                    AS developer,
 COUNT(*)                  AS model_count
FROM ai_developer AS d
JOIN ai AS a
 ON d.id = a.developer_id
GROUP BY d.id, d.name;
```

Result:

developer	model_count
abc Filter...	abc Filter...
OpenAI	3
Google AI	2
DeepMind	1
Stability AI	1
EleutherAI	1
Anthropic	1
Amazon AI	1
Midjourney Inc.	1
DeepSeek AI	1
xAI	1
Alibaba Cloud	1
Perplexity AI	1
Manus AI	1

3. Find developers who have created more than 1 model.

```
SELECT
 d.name                  AS developer,
 COUNT(*)                  AS models
FROM ai_developer AS d
JOIN ai AS a
 ON d.id = a.developer_id
GROUP BY d.id, d.name
HAVING COUNT(*) > 1;
```

Result:

developer	models
Filter...	Filter...
OpenAI	3
Google AI	2

4. List all open-source AI models (and their developers).

```
SELECT
 a.name       AS model,
 d.name       AS developer
FROM ai AS a
JOIN ai_developer AS d
 ON a.developer_id = d.id
WHERE a.open_source = TRUE;
```

Result:

model	developer
BERT	Google AI
Stable Diffusion	Stability AI
Whisper	OpenAI
GPT-NeoX	EleutherAI
DeepSeek-R1	DeepSeek AI
Qwen 3	Alibaba Cloud

5. Show models released between 2018 and 2022, inclusive.

```
SELECT
 name,
 release_year
FROM ai
WHERE release_year BETWEEN 2018 AND 2022
ORDER BY release_year;
```

Result:

name	release_year
abc Filter...	abc Filter...
BERT	2018
DALL·E	2021
ChatGPT	2022
Stable Diffusion	2022
Whisper	2022
GPT-NeoX	2022
Midjourney	2022
Perplexity	2022

6. Find all models whose name contains “GPT” (case-insensitive).

```
SELECT
 name,
 version
FROM ai
WHERE name LIKE '%GPT%';
```

Result:

name	version
abc Filter...	abc Filter...
ChatGPT	4.0
GPT-NeoX	20B

7. Find USA-based developers who have built at least one NLP model.

```
SELECT DISTINCT
 d.name        AS developer
FROM ai_developer AS d
JOIN ai AS a
 ON a.developer_id = d.id
WHERE d.country = 'USA'
 AND a.category = 'NLP';
```

Result:

developer
abc Filter...
OpenAI
Google AI
Anthropic
xAI
Perplexity AI

8. List developers who've released a CV model (via a subquery).

```
SELECT
 name
FROM ai_developer
WHERE id IN (
 SELECT developer_id
 FROM ai
 WHERE category = 'CV'
);
```

Result:

name
abc Filter...
OpenAI
Stability AI
Midjourney Inc.

9. for each developer, show their latest (max) release year.

```
SELECT
 d.name                          AS developer,
 (SELECT MAX(a2.release_year)
  FROM ai AS a2
  WHERE a2.developer_id = d.id
 )                               AS latest_release
FROM ai_developer AS d;
```

Result:

developer	latest_release
abc Filter...	abc Filter...
OpenAI	2022
Google AI	2025
DeepMind	2016
Meta AI	NULL
Stability AI	2022
EleutherAI	2022
Anthropic	2023
Microsoft Research	NULL
Amazon AI	2017
Midjourney Inc.	2022
DeepSeek AI	2025
xAI	2023
Alibaba Cloud	2025
Perplexity AI	2022
Google	NULL
Manus AI	2025

10. Show each developer alongside the count of distinct model categories they cover.

```
SELECT
 d.name                  AS developer,
 COUNT(DISTINCT a.category) AS categories_covered
FROM ai_developer AS d
JOIN ai AS a
 ON a.developer_id = d.id
GROUP BY d.id, d.name;
```

Result:

developer	categories_covered
Filter...	Filter...
OpenAI	2
Google AI	2
DeepMind	1
Stability AI	1
EleutherAI	1
Anthropic	1
Amazon AI	1
Midjourney Inc.	1
DeepSeek AI	1
xAI	1
Alibaba Cloud	1
Perplexity AI	1
Manus AI	1

11. For each AI category, list the most recently released model.

```
SELECT
 a.category,
 a.name          AS latest_model,
 a.release_year
FROM ai AS a
WHERE (a.category, a.release_year) IN (
 SELECT
   category,
   MAX(release_year)
 FROM ai
 GROUP BY category
)
ORDER BY a.category;
```

Result:

category	latest_model	release_year
abc Filter...	abc Filter...	abc Filter...
NLP	Qwen 3	2025
CV	Stable Diffusion	2022
CV	Midjourney	2022
AGI	Gemini 2.5	2025
AGI	Manus AI	2025
Other	DeepSeek-R1	2025

--The End--